



Innovations, Applications & Services

Tan Puay Hoe
Nokia Singapore



We have come of age.....

Time to Innovate

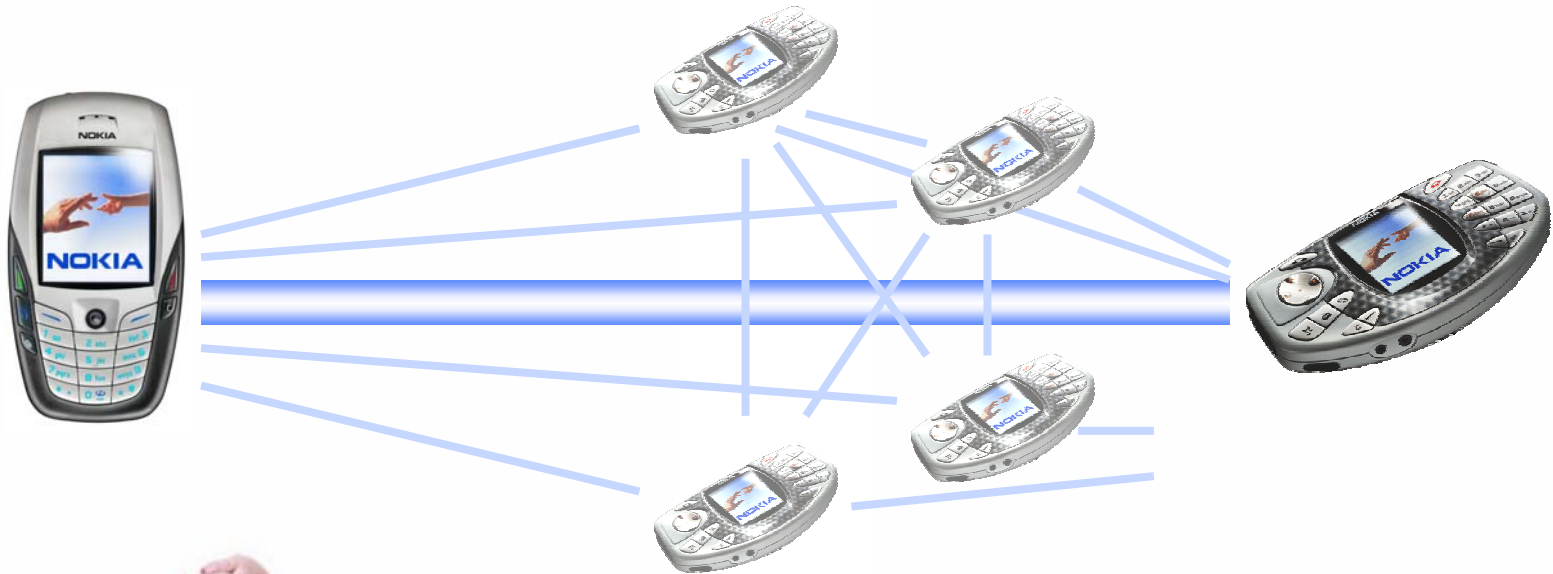
CONTENT

- Role of SIP and IP Multimedia in the Next Generation Services
- SIP Service Examples
- Live Application Use Case Demonstration
- Conclusion and Q&A



Evolving Services need Evolving Service Enablers

**What we can offer in an IP-based world
is something different:**

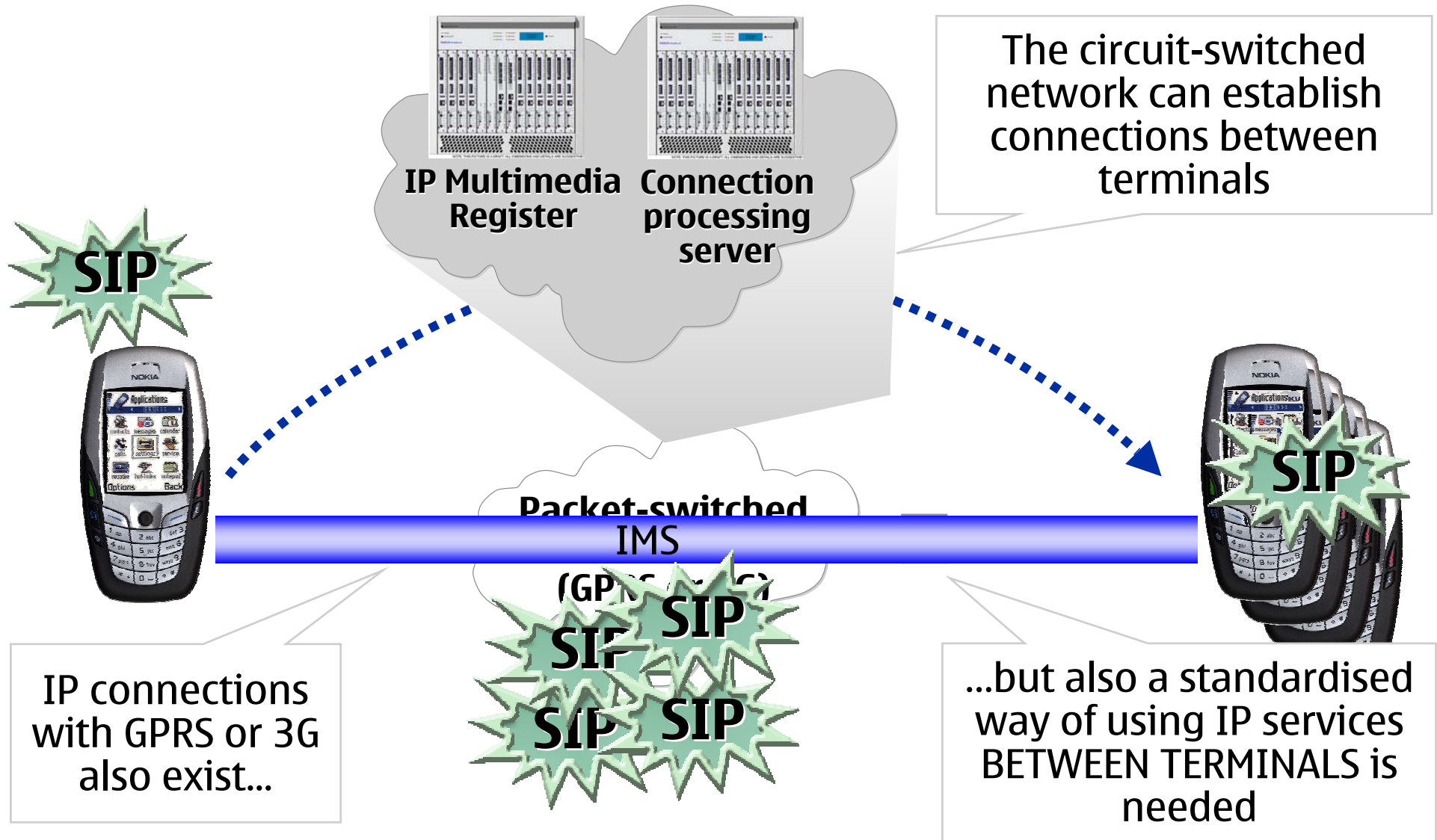


**a direct channel between two terminals
or a group of terminals...**

...and applications running in the terminals



SIP and IP Multimedia Subsystem = Connecting People!

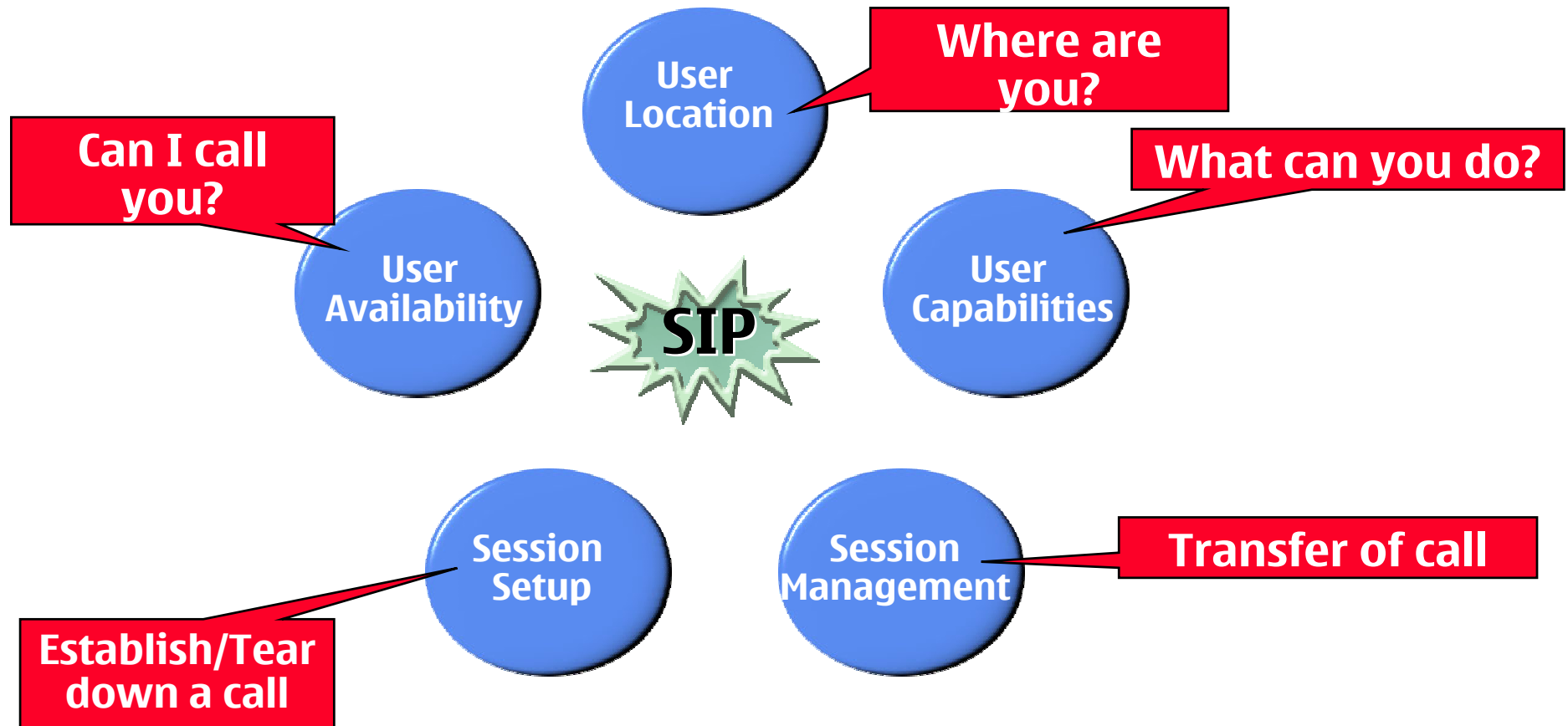


What is SIP...



- **Session Initiation Protocol:**
 - Defined by the Internet Engineering Task Force (IETF);
 - Designed after HTTP.
- SIP is used for initiation of interactive communications sessions between users. SIP also handles termination and modifications of sessions;
- Although originally designed for multimedia sessions:
 - Scope has been extended: presence, gaming, instant messaging, etc.;
 - SIMPLE: **S**IP for **I**ntant **M**essaging and **P**resence **L**everaging **E**xtensions.
- Smarter than your "usual transport":
 - Personal addressing, routing, etc.
- Since its approval in early 1999 as an official standard, SIP has gained wide market acceptance for communications signalling on the Internet;
- Adopted by 3GPP in Release 5;
- More information from IETF RFC-3261.

SIP Covers 5 Aspects of Multimedia Communication



SIP Works with Other Protocols



In addition to SIP there is a need for a number of different standards and protocols to come together to

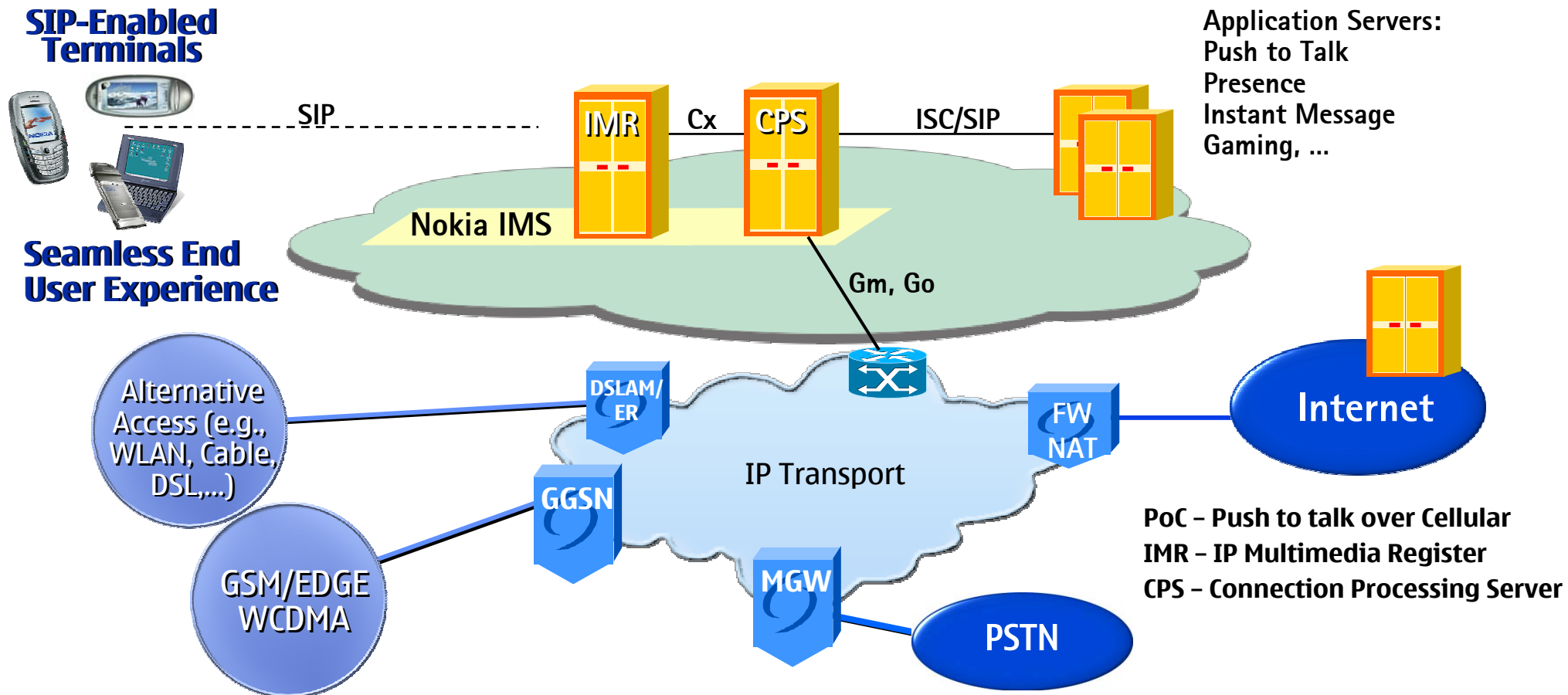
- ensure transport (RTP)
- control delivery of streaming data (RTSP)
- describe multimedia sessions (SDP)
- authenticate users (RADIUS, DIAMETER)
- provide directories (LDAP)
- be able to guarantee voice quality (RSVP)
- inter-work with today's telephone network

SIP Service Creation



- SIP service architecture follows the Internet's successful design.
- Services are developed in the application layer and the network is used exclusively for data transport.
- Services are created with Internet based protocols and languages (e.g. SIP CPL, SIP/CGI, SIP Servlet).
- The benefit is that when all services are created with modular IP protocols, the creation and integration of these services is much faster and easier than of those based on telecom protocols.
- Services are created by operators, 3rd party developers and end users.
- Services and applications can be installed and can run in terminals

Nokia IP Multimedia Subsystem Solution - Architecture



SIP Services

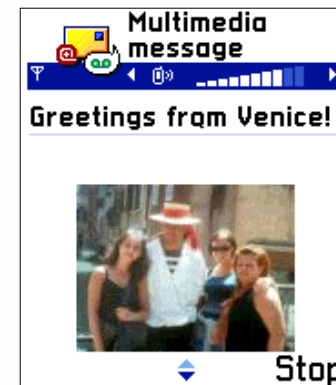
- Voice
- Presence
- Instant messaging
- Gaming
- Rich call
- Conference call
- Event notification
- User Profiling
- Chat
- White boarding, blogging
- Video on Demand
- Interactive-TV
- Combination of services



Voice



Presence



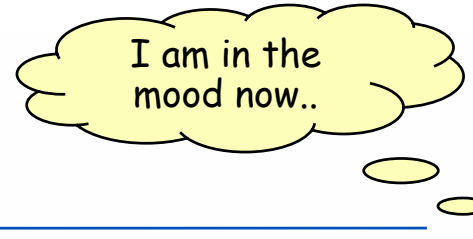
Messaging



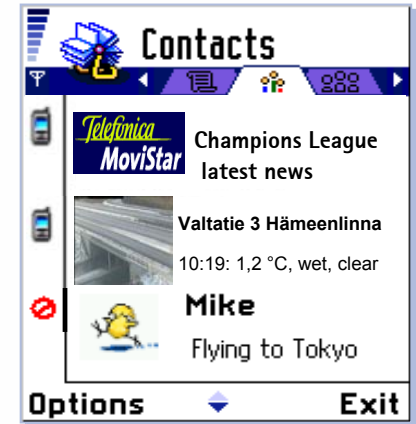
Presence tells you when and how to communicate



'Check first how to communicate'



Presence supports self-expression



Presence can also be used for advertising and information sharing



Presence enables and enriches other services



What is PoC ?

- ★ Push to talk
- ★ Interactive Games
- ★ Content sharing
- ★ Real Time Video Sharing

- PoC = Push to talk over Cellular
- New VoIP service over GPRS networks
- Brings new functionality to cellular phones
 - direct access push-to-talk
 - one-to-one, one-to-many
- Pre-defined groups, ad-hoc groups
- Default, Listened, Do not disturb



1.
**Push to Talk
Key**



2.
**Talk &
Release**

Do you want
to go to
lunch?

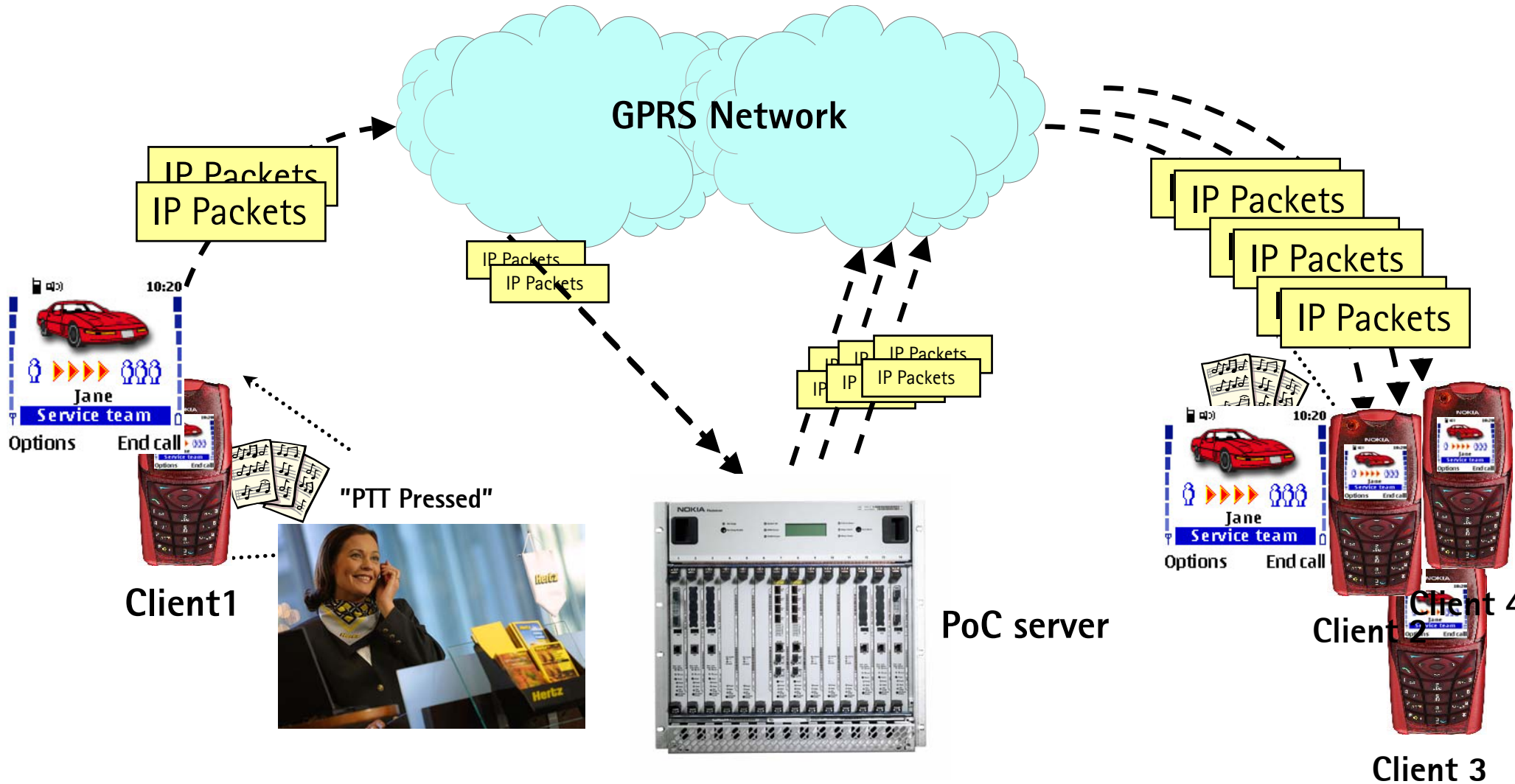
3.

**Sounds
directly in
receiving
end!**



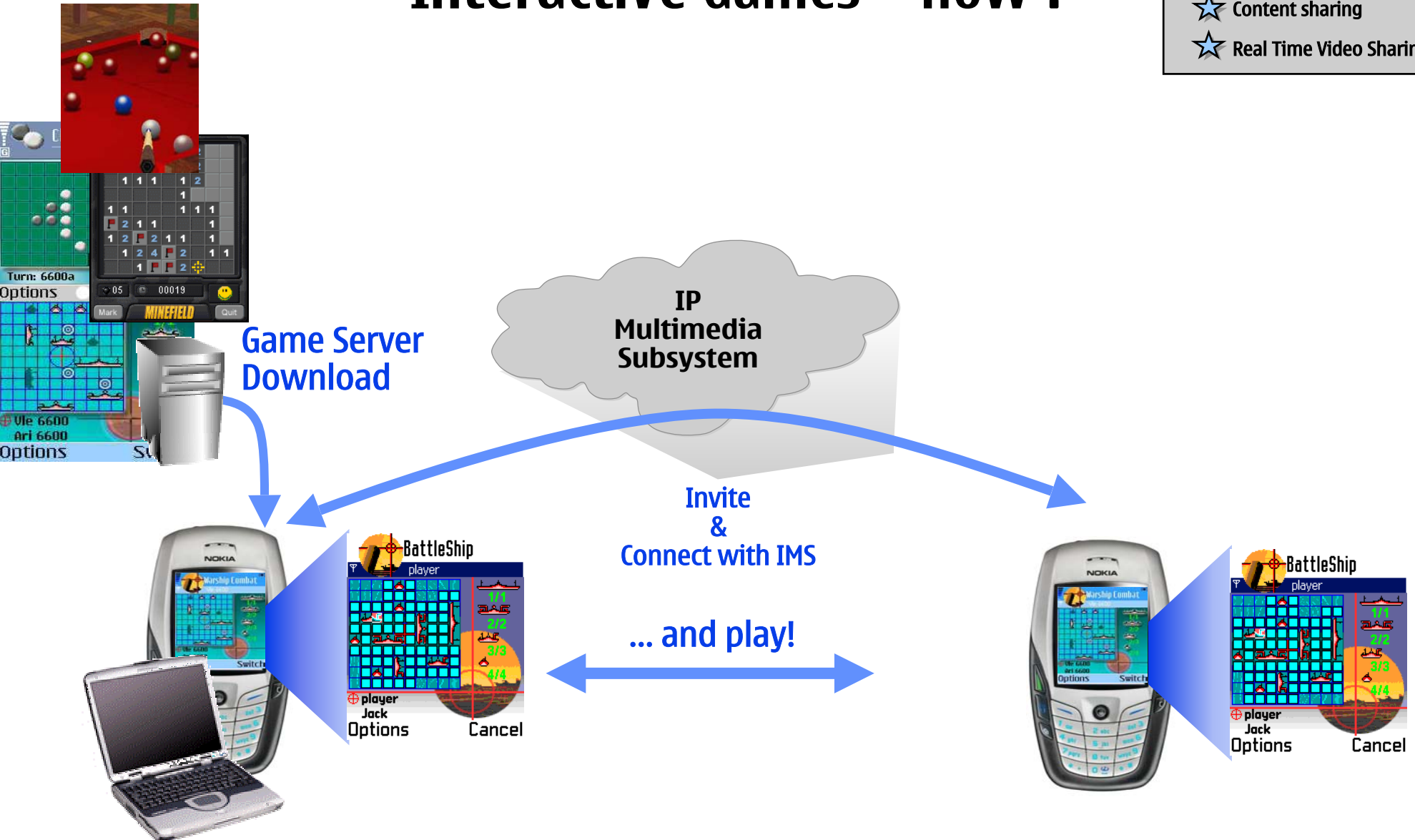
Do you want
to go to
lunch?

Half duplex VoIP Call - PoC



Interactive Games – How ?

- ★ Push to talk
- ★ **Interactive Games**
- ★ Content sharing
- ★ Real Time Video Sharing



Game Use Case Example

1. Luke and Han has started the StarGeeks game. **The terminals are updating the active Game status to the server.** The update will trigger a status change in their presence information related to the game

2. Luke opts for solo gaming and plays offline while Han opts to **offer his availibilty for competition**

3. After the gaming session ends, Luke opts to **report the high scores to the game server**

4. Luke decides to compete and **requests player list** from server for opponent discovery

5. The server returns Luke a list of **available opponents based on some criteria**

6. After selecting Han as the probable opponent, Luke sends Han an **invitation to compete**

7. Han accepts Luke's invitation

8. They start playing and the **game session starts**

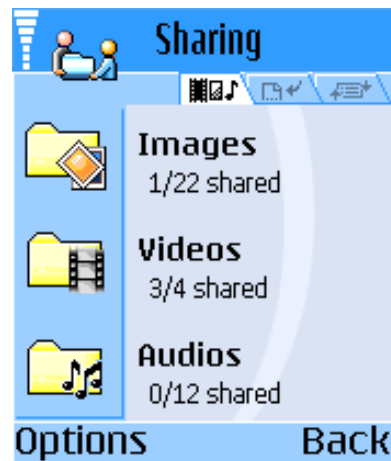
9. After the game session ends, both of them **report scores to the server**



Content Sharing, My Shared Folders

- ★ Push to talk
- ★ Interactive Games
- ★ **Content sharing**
- ★ Real Time Video Sharing

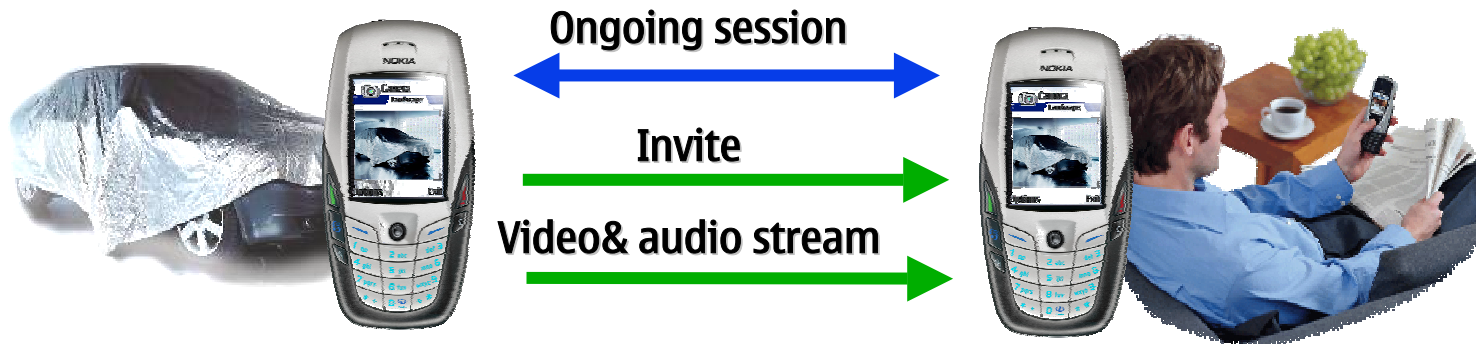
- First, direct file sharing between terminals
- Can be used to transfer files between users or later between group of users
- Sharing also between PC & Mobile



Real-time video sharing

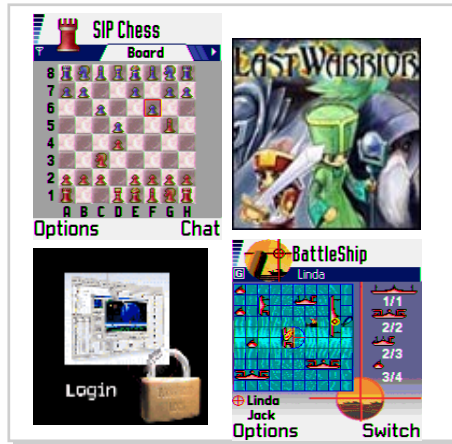
- ★ Push to talk
- ★ Interactive Games
- ★ Content sharing
- ★ **Real Time Video Sharing**

Share the moment instantly in real-time, “Show What You See”

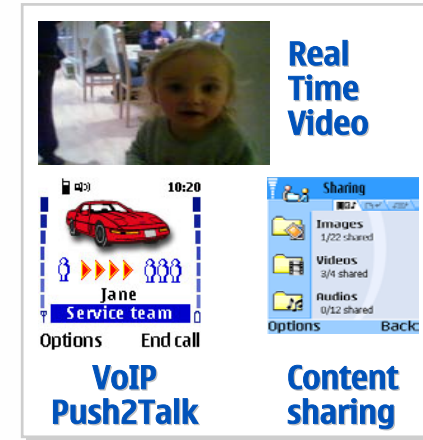


- Enables spontaneous behavior
- Share the camera view or video clip whilst in an ongoing session
- Unidirectional mobile video streaming between peer users
- Enables convergence with PC clients
- Ideal service for WCDMA and EDGE networks

Typical Use Case of IP Multimedia



Interactive applications



Multimedia communication

... all these with a single session control mechanism for better profitability = **IP Multimedia Subsystem**

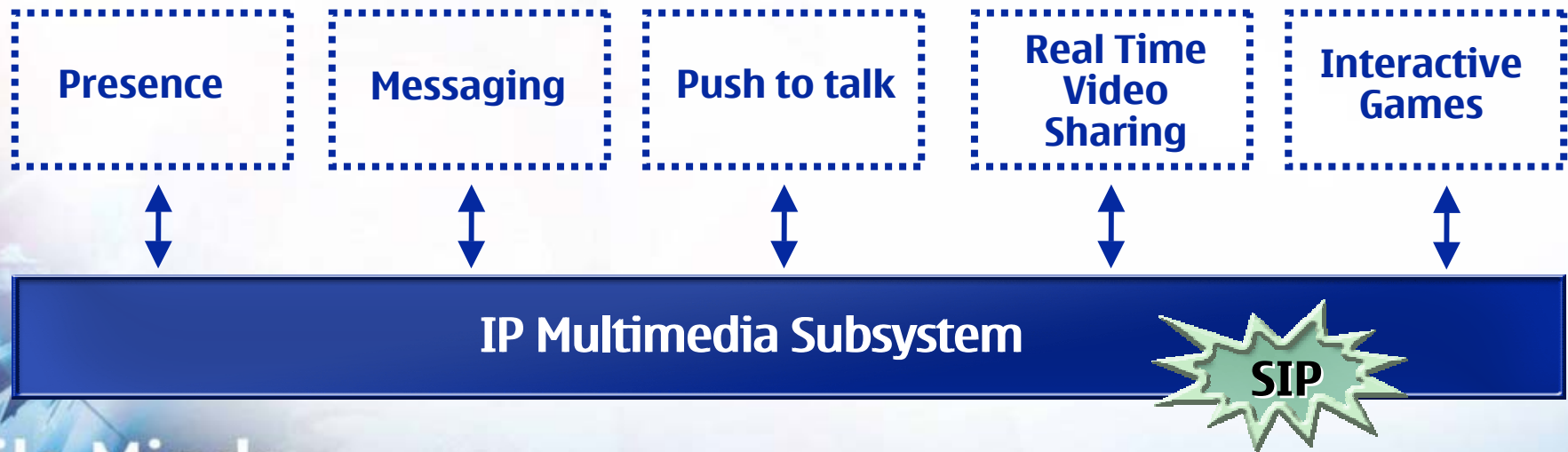


Role of the IP Multimedia Subsystem



SIP provided as a serious network service.

- a clean network architecture, clear standard for interoperability
- build once, use perpetually (avoid building technology silos application by application)
- end user convenience: single address, registration and authentication for all applications
- mobile operators add value on top of plain connectivity by leveraging the synergies with other aspects of the mobile service offering



Conclusions – how & when ?

- Mobile IP Multimedia is happening now and applications can be developed for Nokia Terminals
- At the beginning the SIP stack (& APIs) will be downloadable to terminals, later the Nokia SIP stack will be integrated into Symbian OS
- Since it is a Plug-in, Nokia Series 60 SDK 2.0 for Symbian OS needs to be installed before installing the Plug-in
- Deliverables
 - Installation program
 - User Guide
 - Release Notes
 - Example SIP Application
 - SIP Stack SIS file
 - Documentation:
 - White Paper & FAQ
- SIP Pug in is available via www.forum.nokia.com

Start SIP development today with the SIP plug-in and prepare yourselves for new SIP-based applications business.

Go, Innovate